Session 11

Guiding principles and best practices for data capacity building

Building data capacity

- Not everyone needs to be a data scientist.
- Data capacities in policymaking refer to how well infrastructure, systems, and people can use data effectively for decisions.
- Strengthening these abilities across different teams is essential.
- Complementary skillsets help reinforce the entire data value chain and improve overall decision-making.

1. Identify gaps in data skills and statistical capacities

Types of skills

Technical know-how

Mechanical, ICT, mathematical, scientific and/ or other technology-related skills such as proficiency with a standard statistical package (e.g., R; Stata; SPSS) to carry out data processing, analysis and dissemination.

Work know-how

An employee's knowledge and understanding of work practices, processes, statistical concepts, definitions and classifications, business models, and organisational culture in use in their organisation, as well as about the power relationships within and between organisations.

Problem-solving and creative thinking

The traits that support defining, approaching and analysing a problem in a technically correct manner and from a fresh perspective to devise new ways for carrying out tasks, meeting challenges and solving problems.

Data Capability Framework

	PLAN	COLLECT	DESCRIBE	STORE	ANALYSE	USE	SAVE/DESTROY
Employ data coding and classification principles	□	□	■	■	□	■	■
Integrate data	□	□	■	□	■	■	□
Use data processing methodologies	□	□	□	□	■	□	□
Contribute to data outputs, products or service production	■	□	□	□	□	□	□
Perform exploratory data analysis	□	□	□	□	■	□	□
Conduct business intelligence data analysis	□	□	□	□	■	■	□
Conduct statistical data analysis	□	□	□	□	■	■	□
Conduct specialist data analysis	□	□	□	□	■	■	□
Identify and understand data availability	■	■	□	□	□	□	□
Employ data collection methodology	□	■	□	□	□	□	□
Contribute to data access design	■	■	□	□	□	□	□
Contribute to the sourcing and use of administrative data	■	■	■	□	□	■	□
Understand and contribute to data collection process design	■	■	□	□	□	□	□
Describe and summarise data	□	□	■	□	□	□	□
Understand and apply data editing methods	□	□	■	■	■	□	□
Use data quality assurance measures	□	■	■	□	■	■	□
Identify and evaluate data intelligence	■	■	■	□	□	■	□
Improve data processes/systems/products	■	□	□	■	□	■	■
Identify research questions	■	■	□	□	□	■	■
Apply data governance guidance	■	■	■	■	□	□	■
Value organisational data as assets	□	□	■	■	□	□	■
Employ statistical concepts and methodologies	□	■	■	■	■	■	■
Enable others to use and re-use data	□	□	■	■	□	■	■
Employ data and information management concepts	□	■	■	■	■	■	■
Visualise data	□	□	□	□	□	■	□

Data Maturity Assessment Framework

Topic: Engaging with others

Topic: Having the right data skills and knowledge

Topic: Having the right systems

Topic: Knowing the data you have

Topic: Making decisions with data

Topic: Managing and using data ethically

Topic: Managing your data

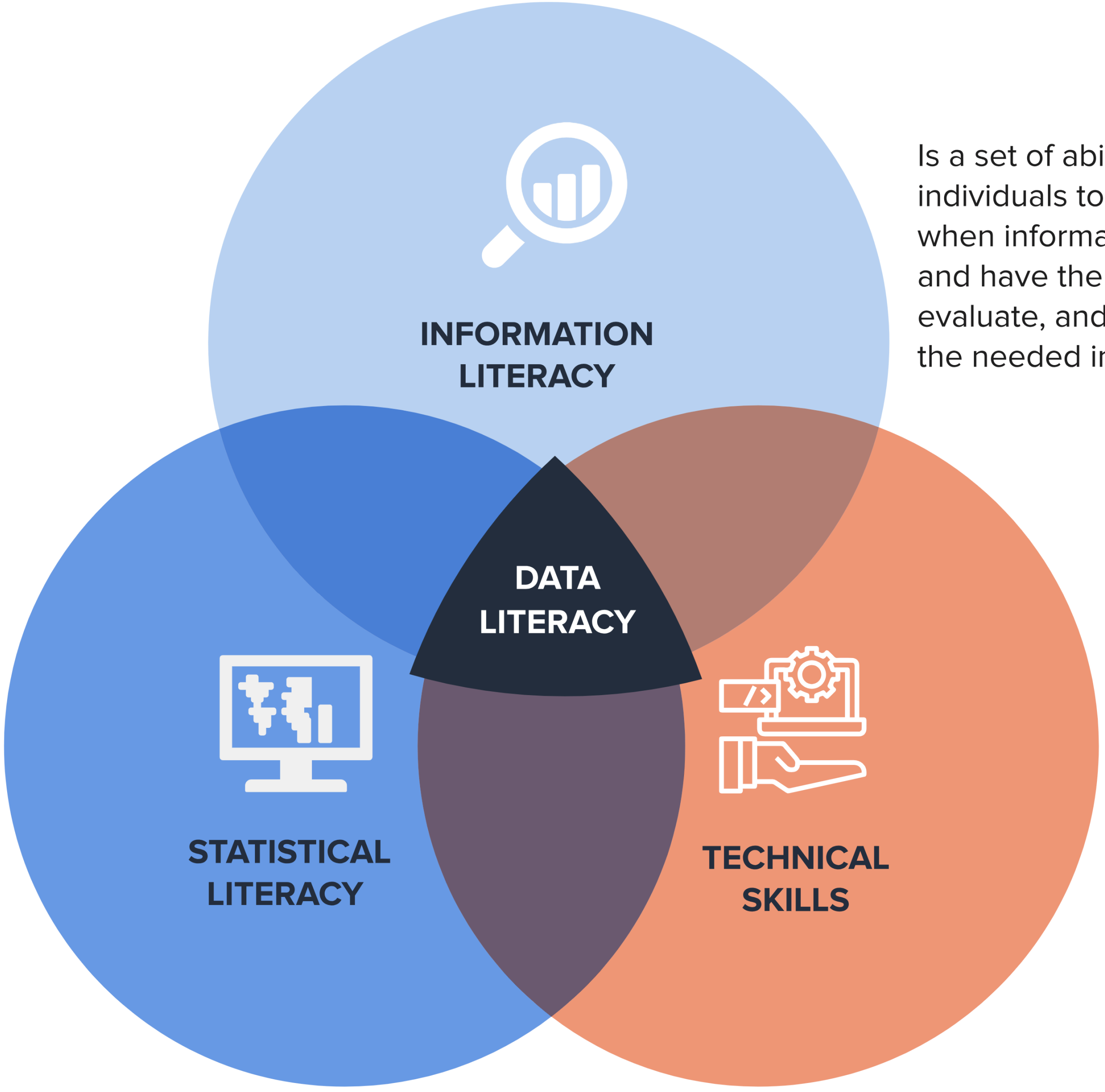
Topic: Protecting your data

Topic: Setting your data direction

Topic: Taking responsibility for data

2. Build data capacities in your team

Data Literacy



Is a set of abilities requiring individuals to “recognize when information is needed and have the ability to locate, evaluate, and use effectively the needed information.

Refers to use of statistics as evidence, encompassing what to count or measure, how to assemble those measurements into summary statistics, what comparisons to form from these statistics and how to communicate these statistics.

Refer to skills that are essential for handling, analyzing, interpreting, and deriving insights from data, which has become increasingly important in the digital age.

Source: UN Data Revolution Group

Data Skills Framework

3. Operationalise data capacity building

Short-term

- Training workshops
- Online training resources
- Knowledge sharing on global innovations and use cases

Medium-term

- Internal data champions
- Periodic data skills training
- Pilot data projects
- Build complementary skills with other stakeholders
- Build on local knowledge and capacities
- Build network of data practitioners across your organization

Long-term

- Embed data skills in job roles
- Institutionalize data training programs
- Partner with think tanks and external organizations
- Partner with academia and local universities